Contents.

[An Asterisk denotes that the Paper is not printed in full.]

Volumes II, III, and IV of the *Cambridge Mathematical Journal* have on the title pages the dates 1841, 1843, 1845 respectively, and volume VIII of the *Cambridge Philosophical Transactions* has the date 1849. As each of these volumes extends over more than a single year, I have added, see p. vii, the year of publication for the papers 1, 2, … 12. In all other cases, the year of publication is shown by the date on the title page of the volume.

Classification.

Geometry

Analysis

1.

ON A THEOREM IN THE GEOMETRY OF POSITION.

[From the Cambridge Mathematical Journal, vol. II. (1841), pp. 267–271.]

We propose to apply the following (new ?) theorem to the solution of two problems in Analytical Geometry.

Let the symbols

$$|\,\alpha\,|, \quad \begin{vmatrix} \alpha, & \beta \\ \alpha', & \beta' \end{vmatrix}, \quad \begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}, \quad \&c.$$

denote the quantities

$$\alpha, \quad \alpha\beta' - \alpha'\beta, \quad \alpha\beta'\gamma'' - \alpha\beta''\gamma' + \alpha'\beta''\gamma - \alpha'\beta\gamma'' + \alpha''\beta\gamma' - \alpha''\beta'\gamma, \quad \&c.$$

(the law of whose formation is tolerably well known, but may be thus expressed,

$$|\,\alpha\,| = \alpha, \qquad \begin{vmatrix} \alpha, & \beta \\ \alpha', & \beta' \end{vmatrix} = \alpha\,|\,\beta'\,| - \alpha'\,|\,\beta\,|,$$

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} = \alpha \begin{vmatrix} \beta', & \gamma' \\ \beta'', & \gamma'' \end{vmatrix} + \alpha' \begin{vmatrix} \beta'', & \gamma'' \\ \beta, & \gamma \end{vmatrix} + \alpha'' \begin{vmatrix} \beta, & \gamma \\ \beta', & \gamma' \end{vmatrix}, \quad \&c.$$

the signs $+$ being used when the number of terms in the side of the square is odd, and $+$ and $-$ alternately when it is even.)

Then the theorem in question is

$$\begin{vmatrix} \rho\alpha + \sigma\beta + \tau\gamma \ldots, & \rho\alpha' + \sigma\beta' + \tau\gamma' \ldots, & \rho\alpha'' + \sigma\beta'' + \tau\gamma'' \ldots \\ \rho'\alpha + \sigma'\beta + \tau'\gamma \ldots, & \rho'\alpha' + \sigma'\beta' + \tau'\gamma' \ldots, & \rho'\alpha'' + \sigma'\beta'' + \tau'\gamma'' \ldots \\ \rho''\alpha + \sigma''\beta + \tau''\gamma \ldots, & \rho''\alpha' + \sigma''\beta' + \tau''\gamma' \ldots, & \rho''\alpha'' + \sigma''\beta'' + \tau''\gamma'' \ldots \end{vmatrix} = \begin{vmatrix} \rho, & \sigma, & \tau \ldots \\ \rho', & \sigma', & \tau' \ldots \\ \rho'', & \sigma'', & \tau'' \ldots \end{vmatrix} \cdot \begin{vmatrix} \alpha, & \beta, & \gamma \ldots \\ \alpha', & \beta', & \gamma' \ldots \\ \alpha'', & \beta'', & \gamma'' \ldots \end{vmatrix}.$$

(This theorem admits of a generalisation which we shall not have occasion to make use of, and which therefore we may notice at another opportunity.)

To find the relation that exists between the distances of five points in space.

We have, in general, whatever x_1, y_1, z_1, w_1, &c. denote

$$\begin{vmatrix} {x_1}^2 + {y_1}^2 + {z_1}^2 + {w_1}^2, & -2x_1, & -2y_1, & -2z_1, & -2w_1, & 1 \\ {x_2}^2 + {y_2}^2 + {z_2}^2 + {w_2}^2, & -2x_2, & -2y_2, & -2z_2, & -2w_2, & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ {x_5}^2 + {y_5}^2 + {z_5}^2 + {w_5}^2, & -2x_5, & -2y_5, & -2z_5, & -2w_5, & 1 \\ 1, & 0, & 0, & 0, & 0, & 0 \end{vmatrix}$$

multiplied into

$$\begin{vmatrix} 1, & x_1, & y_1, & z_1, & w_1, & {x_1}^2 + {y_1}^2 + {z_1}^2 + {w_1}^2 \\ 1, & x_2, & y_2, & z_2, & w_2, & {x_2}^2 + {y_2}^2 + {z_2}^2 + {w_2}^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1, & x_5, & y_5, & z_5, & w_5, & {x_5}^2 + {y_5}^2 + {z_5}^2 + {w_5}^2 \\ 0, & 0, & 0, & 0, & 0, & 1 \end{vmatrix}$$

equals

$$\begin{vmatrix} \overline{x_1-x_1}^2+\overline{y_1-y_1}^2+\overline{z_1-z_1}^2+\overline{w_1-w_1}^2, & \overline{x_1-x_2}^2+\ldots, & \overline{x_1-x_3}^2+\ldots, & \overline{x_1-x_4}^2+\ldots, & \overline{x_1-x_5}^2+\ldots, & 1 \\ \overline{x_2-x_1}^2+\ldots, & \overline{x_2-x_2}^2+\ldots, & \overline{x_2-x_3}^2+\ldots, & \overline{x_2-x_4}^2+\ldots, & \overline{x_2-x_5}^2+\ldots, & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \overline{x_5-x_1}^2+\ldots, & \overline{x_5-x_2}^2+\ldots, & \overline{x_5-x_3}^2+\ldots, & \overline{x_5-x_4}^2+\ldots, & \overline{x_5-x_5}^2+\ldots, & 1 \\ 1, & 1, & 1, & 1, & 1, & 0 \end{vmatrix}.$$

Putting the w's equal to 0, each factor of the first side of the equation vanishes, and therefore in this case the second side of the equation becomes equal to zero. Hence $x_1, y_1, z_1, x_2, y_2, z_2$, &c. being the coordinates of the points 1, 2, &c. situated arbitrarily in space, and $\overline{12}^2, \overline{13}^2$, &c. denoting the squares of the distances between these points, we have immediately the required relation

$$\begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & \overline{15}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & \overline{25}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & \overline{35}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & \overline{45}^2, & 1 \\ \overline{51}^2, & \overline{52}^2, & \overline{53}^2, & \overline{54}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 1, & 0 \end{vmatrix} = 0,$$

which is easily expanded, though from the mere number of terms the process is somewhat long.

Precisely the same investigation is applicable to the case of four points in a plane, or three points in a straight line. Thus the former gives

$$\begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & \overline{14}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & \overline{24}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & \overline{34}^2, & 1 \\ \overline{41}^2, & \overline{42}^2, & \overline{43}^2, & 0, & 1 \\ 1, & 1, & 1, & 1, & 0 \end{vmatrix} = 0;$$

The latter gives

$$\begin{vmatrix} 0, & \overline{12}^2, & \overline{13}^2, & 1 \\ \overline{21}^2, & 0, & \overline{23}^2, & 1 \\ \overline{31}^2, & \overline{32}^2, & 0, & 1 \\ 1, & 1, & 1, & 0 \end{vmatrix} = 0;$$

or expanding,

$$\overline{12}^2 + \overline{13}^2 + \overline{23}^2 - 2\cdot\overline{12}\cdot\overline{13} - 2\cdot\overline{13}\cdot\overline{23} - 2\cdot\overline{12}\cdot\overline{23} = 0;$$

which may be derived immediately from the equation

$$\pm\overline{12} \pm \overline{13} = \pm\overline{23},$$

and is the simplest form under which this equation, cleared of the ambiguous signs, can be put.

(The above result may be deduced so elegantly from the general theory of elimination, that notwithstanding its simplicity it is perhaps worth mentioning.)

Let

$$\overline{x_a - x_b} = \alpha, \qquad \overline{x_b - x_c} = \beta, \qquad \overline{x_c - x_a} = \gamma;$$

then

$$\overline{12}^2 = \gamma^2, \qquad \overline{23}^2 = \alpha^2, \qquad \overline{31}^2 = \beta^2, \qquad \text{and} \quad \alpha+\beta+\gamma = 0;$$

from which α, β, γ are to be eliminated. Multiplying the last equation by $\beta\gamma, \gamma\alpha, \alpha\beta$, and reducing by the three first,

$$\begin{aligned} 0\cdot\alpha + \overline{12}^2\cdot\beta + \overline{31}^2\cdot\gamma + \alpha\beta\gamma &= 0, \\ \overline{12}^2\cdot\alpha + 0\cdot\beta + \overline{23}^2\cdot\gamma + \alpha\beta\gamma &= 0, \\ \overline{31}^2\cdot\alpha + \overline{23}^2\cdot\beta + 0\cdot\gamma + \alpha\beta\gamma &= 0, \\ \alpha+\beta+\gamma+0\cdot\alpha\beta\gamma &= 0; \end{aligned}$$

[Paper 1 continues on the following page (printed p. 4); see next chunk.]